



Sensing microorganisms in the gut triggers the immune response in *Eisenia andrei* earthworms



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ARTICLE INFO

Article history:

Received 6 November 2015

Received in revised form

30 November 2015

Accepted 2 December 2015

Available online 10 December 2015

Keywords:

Earthworm

Mucosal immunity

CCF

Coelomocyte

Microorganism

ABSTRACT

The tube-within-tube body plan of earthworms is appropriate for studying the interactions of microorganisms with the immune system of body cavities such as the digestive tract and coelom. This study aims to describe the immune response on the molecular and cellular level in the coelomic cavity and the gut of the earthworm *Eisenia andrei* after experimental microbial challenge by administering two bacterial strains (*Escherichia coli* and *Bacillus subtilis*) or yeast *Saccharomyces cerevisiae* to the environment. The changes in mRNA levels of defense molecules (pattern recognition receptor CCF, lysozyme, fetidin/lysenins) in the coelomocytes and gut tissue were determined by quantitative PCR. The immune response at a cellular level was captured in histological sections, and the expression of CCF was localized using *in situ* hybridization. Coelomocytes respond to the presence of bacteria in the coelomic cavity by increasing the mRNA levels of defense molecules, especially CCF. The immune response in gut tissue is less affected by microbial stimulation because the epithelial cells of gut exhibit basically strong mRNA synthesis of *ccf* as a defense against the continuous microbial load in the gut lumen. The cellular immune response is mediated by coelomocytes released from the mesenchymal lining of the coelomic cavity. These combined immune mechanisms are necessary for the survival of earthworms in the microbially rich environment of soil.

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1. Introduction

Earthworms represent an appropriate model for the study of the immune system of invertebrates. Earthworms, like other invertebrates, rely generally on the innate immune system, including both cell-mediated and humoral mechanisms (Cooper et al., 2002). The body plan of the earthworm is based on a tube-within-tube concept, with the true coelomic cavity surrounding the gut. Earthworms are cosmopolitan and can be found in all types of soil habitats (Edwards and Bohlen, 1996). From the immunological perspective, soils are perhaps the most complex microbial habitats comprising the broad spectrum of soil bacteria, algae, fungi and protists that exhibit almost endless variations in potential

interactions with earthworms (Drake and Horn, 2007; Zirbes et al., 2012).

The gut surface is in the permanent contact with ingested soil. Moreover, the coelomic cavity is not aseptic (Dales and Kalac, 1992), as each segment of the cavity interfaces with the outer environment via excretion organs, the nephridia, and dorsal pores that enable microorganisms to enter the coelomic cavity (Cameron, 1932). As a consequence, the epithelial surfaces of both the coelom and the gut interact with naturally occurring soil microorganisms. The mesenchymal lining of the coelom is the main source of free coelomocytes (Liebmann, 1942). Part of the coelomocyte population is represented by phagocytic cells, amoebocytes, which are main cellular effectors in the immunological response (Sima, 1994).

In our laboratory, we have focused for the last two decades on researching various defense molecules in the species *Eisenia andrei* (formerly *E. fetida andrei*, NCBI Taxonomy: 168636), namely coelomic cytolytic factor (CCF), antimicrobial proteins with hemolytic function and lysozyme.

Coelomic cytolytic factor has been described as a pattern

Abbreviations: CCF, coelomic cytolytic factor; CFU, colony forming unit; DIG, digoxigenin; DTT, dithiothreitol; LBP, lipopolysaccharide-binding protein; PAMPs, pathogen-associated molecular patterns; PRR, pattern recognition receptor; SSC, saline-sodium citrate buffer; TLR, Toll-like receptor.

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recognition receptor (Beschin et al., 1998). Moreover, CCF acts as an opsonin in the coelomic fluid (Bilej et al., 1995). The range of recognition of various pathogen-associated molecular patterns (PAMPs) is given by the presence of two distinct lectin-like domains in the CCF molecule, allowing the recognition of yeast, Gram-negative bacteria and Gram-positive bacteria (Bilej et al., 2001). It was also shown that CCF triggers the activation of the phenoloxidase cascade (Beschin et al., 1998), which is a basic defense mechanism in many invertebrate species. Further, the mRNA level of CCF in coelomocytes was shown to be upregulated upon the injection of microorganisms into the coelomic cavity (Köhlerova et al., 2004).

Antimicrobial proteins with hemolytic activity were described for the first time in the late 1960s (Du Pasquier and Duprat, 1968) and later characterized as two glycoproteins with molecular masses of 40 and 45 kDa (Roch et al., 1981). The protein isoforms were named fetidins, and the gene coding for the 40 kDa molecule was sequenced (Lassegues et al., 1997; Milochau et al., 1997). Independently, a similar protein secreted by coelomocytes was purified from coelomic fluid and named lysenin (Sekizawa et al., 1996). Simultaneously, other lysenin-like proteins were described (Sekizawa et al., 1997). It was later confirmed that fetidin and lysenin represent two distinct molecules encoded by two different genes (Prochazkova et al., 2006). Changes in the composition of the microbial environment result in variations of fetidin and lysenins mRNA levels in the coelomocytes of *E. andrei* (Dvorak et al., 2013).

Lysozyme is an evolutionarily conserved antimicrobial protein. The *E. andrei* lysozyme shares high homology with other invertebrate lysozymes (Joskova et al., 2009).

In this study, we focused on the response of the immune system of *E. andrei* earthworms after experimental microbial stimulations with Gram-positive and Gram-negative bacterial strains and yeasts. The exposure of earthworms to high microbial loads resulted in increased numbers of microbes in the gut and especially in the coelomic cavity. The comparison of mRNA levels of selected molecules (CCF, fetidin/lysenins, lysozyme) in the coelomocytes and in the gut revealed a particularly strong change in the expression of the CCF molecule. We found that coelomocytes exhibit a significantly higher change in CCF expression compared to the gut tissue upon microbial stimulations. On the other hand, the midgut exhibits the highest constitutive expression of CCF in the earthworm body. We suggest that the experimental microbial challenge stimulates the immune response in the gut. This response triggers the increased release of phagocytic coelomocytes from the mesenchymal lining of the coelom and thus increases the defense reaction in the coelomic cavity of earthworms.

2. Materials and methods

2.1. Microbial stimulation of earthworms

Earthworms were kept on moist paper towels without food for three days to decrease the number of microorganisms in the coelom and to empty the gut content. The animals were then transferred to Petri dishes containing paper towels soaked with a suspension of specific microorganisms diluted in modified isotonic PBS (diluted with water 3:2 v/v, pH 7.3). *Escherichia coli* O55, *Bacillus subtilis* W23 and *Saccharomyces cerevisiae* S288 (all microorganisms were obtained from the Institute of Microbiology of the CAS) in a total number of 9×10^9 Colony Forming Units (CFUs) of particular microorganisms were used for the stimulation of earthworms. Earthworms were maintained in the environment with the microorganisms for 1–7 days at RT. Non-stimulated earthworms were kept in pure modified isotonic PBS and used as a negative control.

2.2. Harvesting of coelomic fluid and isolation of gut tissue, CFU estimation

Coelomic fluid containing free coelomocytes was harvested by puncturing the coelomic cavity with a Pasteur micropipette. Coelomocytes were collected by subsequent centrifugation (10 min, $500 \times g$, 4 °C). Gut tissue was dissected from the post-clitellar body part rinsed in modified PBS solution and homogenized using manual homogenizer Squisher™ (Zymo Research) in 50 µl of modified isotonic PBS. The number of CFUs in cell-free coelomic fluid and in gut homogenate was estimated by a plate counting method at each time point for both microbial stimulations and negative controls. Microorganisms were incubated on LB plates supplemented with 2% dextrose at 30 °C.

2.3. RNA isolation, reverse transcription, quantitative PCR

Coelomocytes and gut tissue samples were pooled separately from five earthworms respectively, and total RNA was isolated using Direct-zol RNA Miniprep (Zymo Research) according to the manufacturer's protocol. The concentration and quality of RNA were checked by Nanodrop 1000 and denaturing RNA electrophoresis. Samples were treated with DNase I (Fermentas), and 500 ng of total RNA was reverse transcribed using oligo(dT)₂₀ primers and SuperScript III Reverse Transcriptase (Life Technologies). Quantitative PCR experiments were performed to determine changes in the mRNA levels of CCF, fetidin/lysenins and lysozyme. To amplify all variants of fetidin, lysenin and lysenin-related genes, universal primers were designed based on conserved regions of these genes (Dvorak et al., 2013). The reaction (total volume 25 µl) was prepared as follows: 12.5 µl iQ™ SYBR® Green Supermix (Bio-Rad), 1 µl of both forward and reverse primers (Table 1) and 4 µl of cDNA diluted in RNase free water (1:20). PCR was performed using a CFX96 Touch™ Real-time PCR Detection System (Bio-Rad). The amplification protocol was as follows: 3 min at 95 °C followed by 40 cycles of 94 °C for 30 s, 59 °C for 40 s and 72 °C for 60 s. The temperature was then gradually increased to 95 °C to obtain melting curves of the amplified fragments, which confirmed the specificity and uniformity of the PCR products. Serial dilutions of cDNA (1:5) were used in qPCR for each primer pair to determine the efficiency of amplification. Changes in gene expression were calculated according to the $2^{-\Delta\Delta C_T}$ (Livak) method. Two reference genes (RPL13, RPL17) were selected as internal controls for the normalization of the expression of other genes. The fold change of mRNA level was related to the change in the negative controls. The results were expressed as the mean ($\pm$ SD) of the values obtained in three independent experiments. Evidence of significant changes was evaluated using the one-sample t-test in the GraphPad Prism software.

2.4. Preparation of RNA probes for in situ hybridization

Probes for *in situ* hybridization were designed to localize the mRNA of CCF (GenBank: AF030028). The 425 bp long DNA template was synthesized by PCR (initial denaturation 94 °C 3 min; 40 repeats of 94 °C 25 s, 60 °C 30 s, 72 °C 45 s; and final extension of 72 °C 7 min) using forward and reverse primers containing sequences for the SP6 or T7 promoters, respectively (Table 1). For the synthesis of DIG-labeled RNA probes, a DIG RNA labeling kit SP6/T7 (Roche) was used according to the manufacturer's protocol.

2.5. In situ hybridization

Fresh pre-clitellar, clitellar and post-clitellar earthworm body parts were embedded in Tissue-Tek (Sakura) and immediately snap

Table 1

List of primers used in quantitative PCR experiments and oligonucleotides CCF T7 and CCF SP6 used for the synthesis of RNA probes.

Name	Sequence
RPL13 for	CACAATTGGAATTGCTGTCG
RPL13 rev	GTGGCATCACCTTGTTAGG
RPL17 for	GCAGAATTCAAGGGACTGGA
RPL17 rev	CTCCTTCTCGGACAGGATGA
CCF for	CTTCACCGACTGGGATCAAT
CCF rev	CGTTGTTGTCGTATTCGTG
Fetidin/lysine for	TGGCCAGCTGCAACTCTT
Fetidin/lysine rev	CCAGCGCTGTTTCGGATTAT
Lysozyme for	GGAACAAGGTGAAGCAGTG
Lysozyme rev	GGCAGTCCAGGTGGTAGTGT
CCF T7	GAAATTAATACGACTACTATAGGGAGAAAGAACCTCAGTCCGTTGGG
CCF SP6	GAAGATTTAGGTGACACTATAGAAGAGACGACAAACACGGATGCTA

frozen in an isopentane bath cooled with liquid nitrogen. Ten micrometer thick transversal sections were prepared using a cryostat at -20°C and then transferred onto a Superfrost Plus Gold slide (Menzel-Gläser, Thermo Scientific). The slides were air dried for 5 min at room temperature and fixed in fresh RNase-free 4% paraformaldehyde (pH 7.2) on ice for 15 min. Sections were digested with 5 $\mu\text{g}/\text{ml}$ proteinase K (Life Technologies) for 2 min and postfixed with 4% paraformaldehyde (pH 6.5) on ice for 8 min. The acetylation step was performed in a solution of 0.25% acetic anhydride in 0.1 M triethanolamine (pH 8) for 10 min. Slides were washed in $2 \times \text{SSC}$ (saline-sodium citrate buffer) and prehybridized for 2 h at 42°C in prehybridization mixture (40% formamide, $4 \times \text{SSC}$, 10% dextran sulfate, $1 \times \text{Denhardt's}$ solution, 10 mM DTT (dithiotreitol), 0.1 mg/ml yeast tRNA). Antisense and sense probes (final concentration 1 ng/ μl) were heat denatured for 10 min at 70°C in the prehybridization mixture and then kept on ice. Sections were hybridized with probes for 18 h at 52°C in a moist chamber. Samples were washed with $4 \times \text{SSC}$ and treated with RNase A for 30 min at 37°C . For the removal of unhybridized probes, a decreasing concentration gradient of SSC buffer at room temperature ($2 \times \text{SSC}$, $1 \times \text{SSC}$, $0.5 \times \text{SSC}$) was used, followed by a final wash step in $0.1 \times \text{SSC}$ at 52°C . Slides were treated with a blocking reagent (Boehringer) for 1 h and then incubated with anti-DIG-alkaline phosphatase antibody (1:5000, Roche) overnight at 4°C . The labeling was visualized using a mixture of 0.04 mM nitroblue tetrazolium, 0.05 mM bromochloroindolyl phosphate and 1.1 mM levamisole in DIG buffer (pH 9.5). Sections were counterstained with a 0.1% nuclear fast red aluminum sulfate solution for 20 min, followed by washing in water and then mounting in Apathy's mounting medium.

2.6. Histology

2.6.1. Hematoxylin/eosin staining

Earthworms stimulated with the microorganisms were examined by using histological techniques. Post-clitellar body parts were fixed in 4% formalin overnight and embedded in paraffin. Transverse sections (5 μm thick) were prepared using a microtome and dried overnight. Sections were deparaffinized in xylene and rehydrated in a decreasing concentration gradient of ethanol. Staining with hematoxylin/eosin was performed according to the protocol (Kiernan, 2008).

2.6.2. Toluidine blue staining

Deparaffinized dehydrated sections were stained in a fresh solution of toluidine blue (final concentration 0.1%) for 20 min. Slides were washed several times in water and then dried overnight. Sections were dehydrated in 96% ethanol, subsequently cleared in

xylene and mounted in Permount medium under cover glass.

3. Results and discussion

3.1. Effect of microbial challenge on the change in number of bacteria in the coelomic cavity and the gut

Earthworms were kept on moist paper towels for a period of three days to minimize the number of microorganisms in the coelom and to empty the gut. This group was used as a negative control. The numbers of bacteria in the coelomic fluid and gut homogenate of non-stimulated earthworms were estimated. Even in the absence of soil, earthworms still possess a significant population of microorganisms in the coelom and in the alimentary tract. The number of bacteria in the coelomic fluid was approximately 5.81×10^5 per milliliter. The number of bacteria in gut homogenate was approximately 3.69×10^6 per milliliter, more than six times higher amount than in the coelomic fluid. The numbers of bacteria in the coelom are kept under control by various mechanisms such as antimicrobial factors (Joskova et al., 2009; Lassegues et al., 1989; Prochazkova et al., 2006) and phagocytic cells (Dales and Kalac, 1992). To confirm the presence of bacteria in the coelomic cavity *in situ*, histological sections were stained with toluidine blue (Fig. 1-I). Bacteria appear as clusters of dark blue dots in the space of the coelom.

In earthworms subjected to high microbial load, we observed increased numbers of microorganisms in both the coelom and the gut (Fig. 1-II). The relative fold change of CFU in the coelomic fluid reached a maximum on the third day after the administration of *E. coli* O55, *B. subtilis* W23, and *S. cerevisiae* S288 into the environment. *E. coli* and *S. cerevisiae* stimulations showed the highest impact on the increase of CFUs in the coelomic fluid (150- and 330-fold increases, respectively). Stimulation with *B. subtilis* affected the number of CFUs in the coelomic fluid to lesser extent (6.5-fold increase). This result can be explained by the presence of antimicrobial proteins, such as lysozyme (Joskova et al., 2009), that inhibit the growth of Gram-positive bacteria in the coelomic fluid. An increased relative fold change of CFU in the gut was detected, similarly to the coelomic fluid, after three days of *E. coli*, *B. subtilis*, and *S. cerevisiae* challenge. However, the increase in the number of CFUs in the gut is much lower in comparison with the coelomic fluid. The range of change varies between twenty to thirty-five-fold higher in comparison with naïve worms. The populations of microorganisms in the gut appear to remain less affected after microbial administration compared to the populations in the coelomic fluid. This result should be attributed to the higher basal number of microorganisms in the digestive tract suggested to be tightly associated with the intestinal wall of earthworms (Singleton et al.,

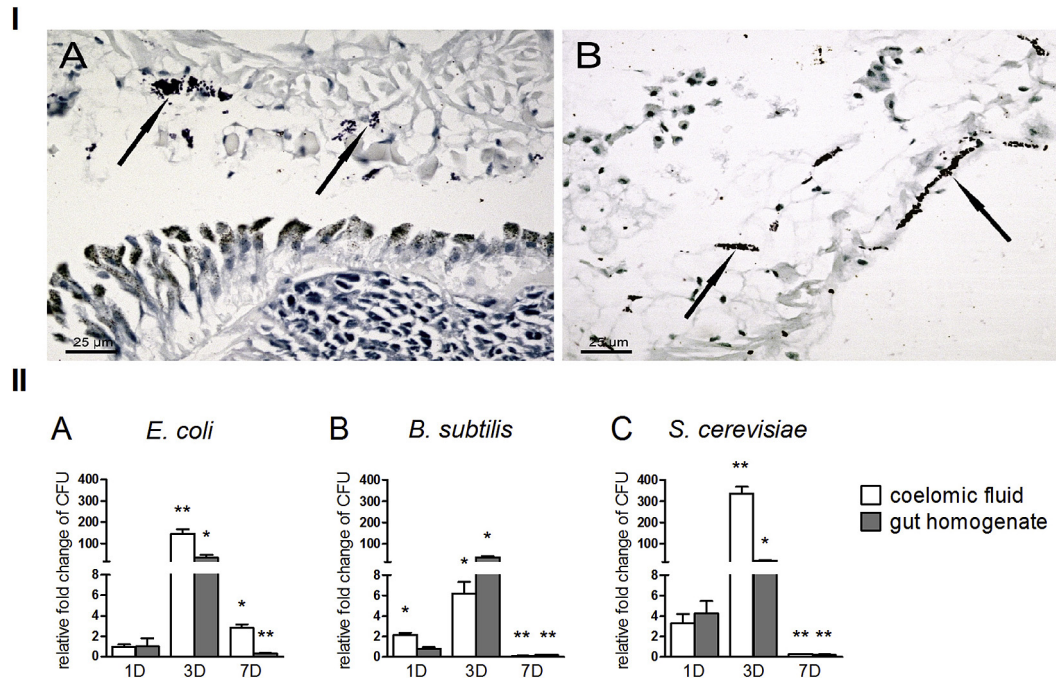


Fig. 1. Changes in number of CFUs in the coelomic cavity and the gut. I. Bacteria were detected in the coelomic cavity in the transversal sections of earthworms. Slides represent sections of animals on day 3 after the microbial challenge. Chloragogen layer and free coelomocytes are shown in Fig. 1A. The coelomic cavity with its floating coelomocytes is shown in Fig. 1B. Clusters of bacteria are marked by arrows. II: Relative fold change of CFUs in earthworms stimulated with *E. coli* (A), *B. subtilis* (B), and *S. cerevisiae* (C) at 1, 3 and 7 days after microbial administration in the coelomic fluid and gut tissue. Differences in CFU number are expressed as fold changes relative to CFU number of non-stimulated animals. Values are means of three experiments ($\pm$ SD) performed in duplicates (* significant at $p < 0.05$, ** significant at $p < 0.01$).

2003). The question of real gut symbiosis in earthworms remains unresolved (Curry and Schmidt, 2007), although bacteria associated with the gut wall of earthworms have been described as determined mainly by the ecological group (Thakuria et al., 2010).

3.2. Effect of microbial challenge on the change in mRNA level of defense molecules in coelomocytes and in the gut

We followed the changes in mRNA levels of pattern recognition receptor CCF and two antimicrobial molecules, fetidin/lysenins and lysozyme, in coelomocytes and in the gut (Fig. 2). We measured the expression up to 7 days after microbial challenge and compared the mRNA levels of challenged and naïve worms. Quantitative PCR of cDNA samples isolated from coelomocytes revealed an increase in the mRNA level of CCF following challenge with all three types of microorganisms. A significant increase in mRNA levels was detected in all three stimulation experiments after one day of microbial challenge. The maximum mRNA levels were observed three days after *E. coli* and *S. cerevisiae* administration, with normalized increases of 30- and 8-fold, respectively. The mRNA levels of fetidin/lysenins were observed to be significantly increased up to three days after all three types of microbial stimulation. It was previously reported that the microbial environment affects the mRNA levels of fetidin and lysenins in the coelomocytes of *E. andrei* earthworms (Dvořák et al., 2013). Moreover, the injection of bacteria into the coelomic cavity has resulted in increased hemolytic activity in the coelomic fluid (Köhlerová et al., 2004). The highest lysozyme mRNA levels were detected after one day of the *B. subtilis* challenge, which is consistent with the results of Joskova et al. (Joskova et al., 2009).

In the gut (Fig. 2), the mRNA levels of CCF were slightly increased one day after the microbial challenge (2-fold increase in the case of all three microbial stimulations) and decreased to the basic mRNA level of naïve worms in the case of *E. coli* and

S. cerevisiae challenge within the next two days. Increases in anti-microbial molecules were detected in the first three days of microbial stimulation in the case of fetidin/lysenins, whereas lysozyme mRNA levels were slightly increased three days post-administration of Gram-positive and Gram-negative bacteria.

The transcription levels of defense molecules in the gut are less affected by microbial challenge compared to coelomocytes. The earthworm digestive tract represents a site of permanent close contact with the microorganisms present in ingested soil (Brown and Doube, 2004; Koubova et al., 2015; Singh et al., 2015). Therefore, this continuous microbial pressure may result in higher basal transcription levels of defense molecules in the gut epithelium than in other tissues.

3.3. In situ hybridization

In situ hybridization showed CCF expression in epithelial cells of the foregut and midgut of *E. andrei* (Fig. 3). CCF mRNA was not detected in the anterior region of the alimentary tract (data not shown). The signal of CCF hybridization is indicated in the clitellar section of the gut (Fig. 3A); however, the dominant expression was observed in the post-clitellar midgut (Fig. 3C). No other tissues or structures exhibited CCF expression. Sense probes used as negative controls did not indicate any unspecific staining in transverse sections. This result is in good agreement with previous findings that the transcription of CCF is localized exclusively in gut tissue and coelomocytes (Skanta et al., 2016). The gene expression of other PRRs has been described in *E. andrei*, such as Toll-like receptor (TLR), which exhibits high gene expression in the gastrointestinal tract, especially in the intestine (Skanta et al., 2013). Another *E. andrei* PRRs lipopolysaccharide-binding protein (LBP) was described to be highly expressed in the anterior part of the digestive tract (Skanta et al., 2016). The expression of CCF in the gut

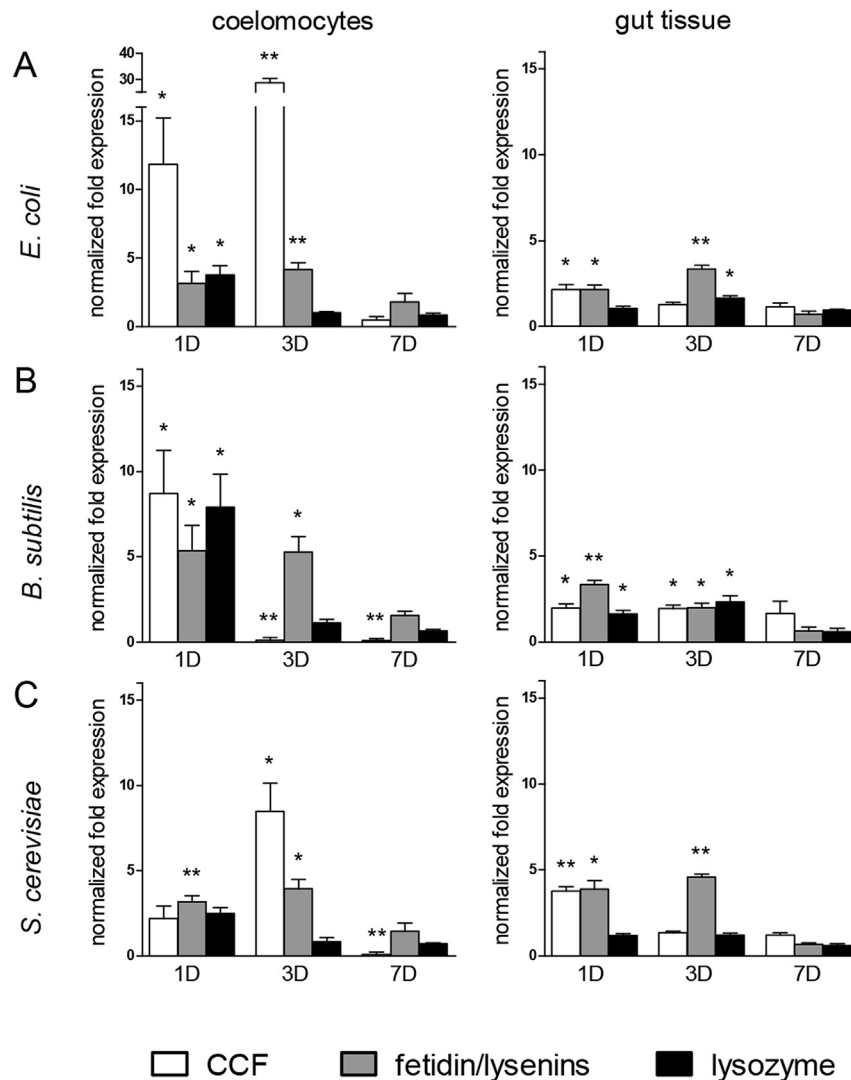


Fig. 2. Changes in mRNA levels of CCF, fetidin/lysenins and lysozyme in coelomocytes and in gut tissue of earthworms stimulated with *E. coli* (A), *B. subtilis* (B), and *S. cerevisiae* (C) at 1, 3 and 7 days after microbial challenge. The mRNA levels were determined by quantitative PCR and normalized to two reference genes (RPL13, RPL17). The values represent fold changes relative to non-stimulated controls. Values are means of three experiments ($\pm$ SD) performed in duplicates (* significant at $p < 0.05$, ** significant at $p < 0.01$).

epithelium is very strong and stable, which confirms the key role of PRRs in the maintenance of host-microbial interactions in the intestinal mucosa (Fukata and Arditi, 2013). The broad recognition specificity of CCF (Bilej et al., 2001) can provide a universal mechanism for the control of diverse communities of gut microorganisms.

3.4. Histology

The hematoxylin/eosin staining procedure was used to obtain a representative histological view and to characterize the intestinal tract structure. The general view of the midgut anatomy is shown in Fig. 4A. The alimentary canal is surrounded by the coelomic cavity. The inner space of the gut is lined with enterocytes with digestive functions. The typhlosolis represents gut invagination that increases the capacity for food digestion and absorption. The chloragogen tissue is attached to the intestine and serves as a center for the synthesis and storage of glycogen and lipids and the excretion of waste products of cell catabolism. Moreover, the chloragogen layer represents the origin of one type of coelomocytes named eleocytes with characteristic autofluorescent vacuoles –

chloragosomes. Another population of coelomocytes – amoebocytes is derived from the mesenchymal lining of the coelomic cavity (Plytycz and Morgan, 2011). Coelomocytes are released to the coelom (Fig. 4B), where they perform either nutritive (eleocytes) or immune (amoebocytes) functions according to their type (Sima, 1994). The mechanism of coelomocyte release was captured in histological sections and is shown in Fig. 4C and in detail in Fig. 4D. It was shown that precursor cells in the coelomic lining proliferate immediately after antigenic stimulation (Bilej et al., 1994). These cells then differentiate and enter the coelomic cavity to fulfill their defense function. However, free coelomocytes have been reported to require repeated contact with stimulating agents to trigger the proliferation response (Bilej et al., 1992). Experimental coelomocyte depletion was shown to be followed by extensive cell proliferation in the coelomic lining (Homa et al., 2008; Klimek et al., 2012). *In vivo* dermal exposure to different immunostimulants induced changes in the number, composition and activity of coelomocytes. In earthworms treated with stimulants, a reduction in the percentage of nutritive eleocytes and an increase in the percentage of immune effector cells, amoebocytes, were observed (Homa et al., 2013). Moreover, it was reported that coelomocytes

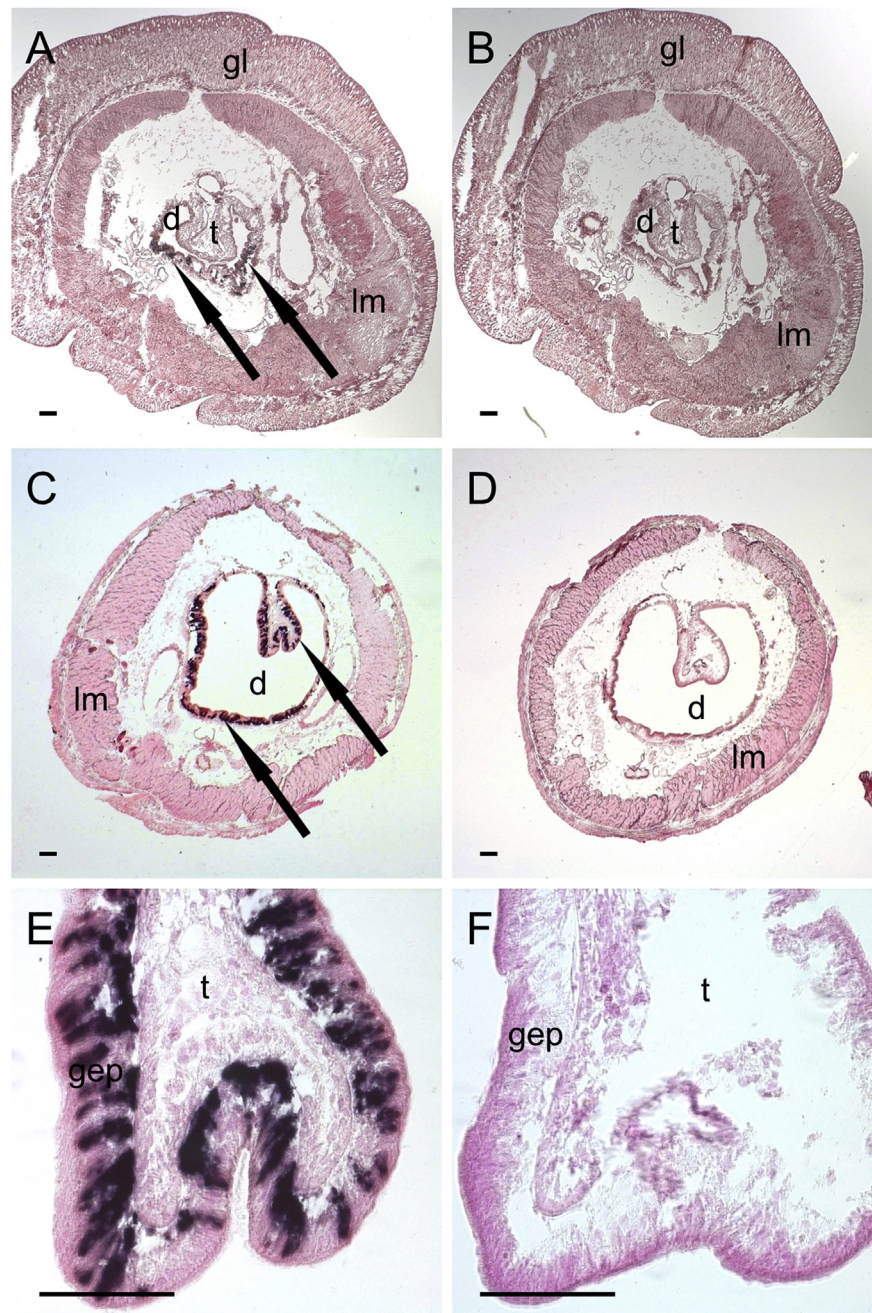


Fig. 3. *In situ* hybridization of CCF in transversal sections of different body parts of earthworm *E. andrei*. The specific signal was detected in the clitellar (A, B) and post-clitellar (C, D; in detail E, F) regions. The positive signals obtained by hybridization with antisense probe are indicated in A, C, E. The reaction for sense probe (negative controls) is shown in B, D, F. Sites of gene expression are marked by arrows. Abbreviations: **d** digestive tract, **gep** gut epithelium, **gl** glandular layer, **lm** longitudinal muscles, **t** typhlosolis. Scale bar = 100 μ m.

respond by directional migration to both bacterial and tissue antigens (Marks et al., 1979). We hypothesize that an increased number of bacteria in the gut triggers signaling pathways that induce cell proliferation and differentiation in the coelomic lining. These cells then enter the coelomic cavity and migrate to the site of infection.

4. Conclusions

Earthworms live in microbially rich habitats. To survive in such environments, they have had to evolve efficient cellular and humoral immune mechanisms. The gut lumen is in direct contact with

the microorganisms present in ingested soil. Similarly, the coelomic cavity is in contact with the outer environment via excretion organs and dorsal pores. In non-stimulated earthworms, the number of bacteria is almost one order of magnitude higher in the gut than in the coelomic fluid. Therefore, it is not surprising that we detected very high constitutional expression of CCF, a pattern recognition receptor with very broad specificity, in the gut. Upon experimental microbial challenge, the increase of transcription of all the observed defense molecules (CCF, fetidin/lysenins, lysozyme) was significantly higher in the coelomocytes than in the gut. Earthworm coelomocytes have been described as exhibiting low proliferation activity. Therefore, the main source of these cells is their release

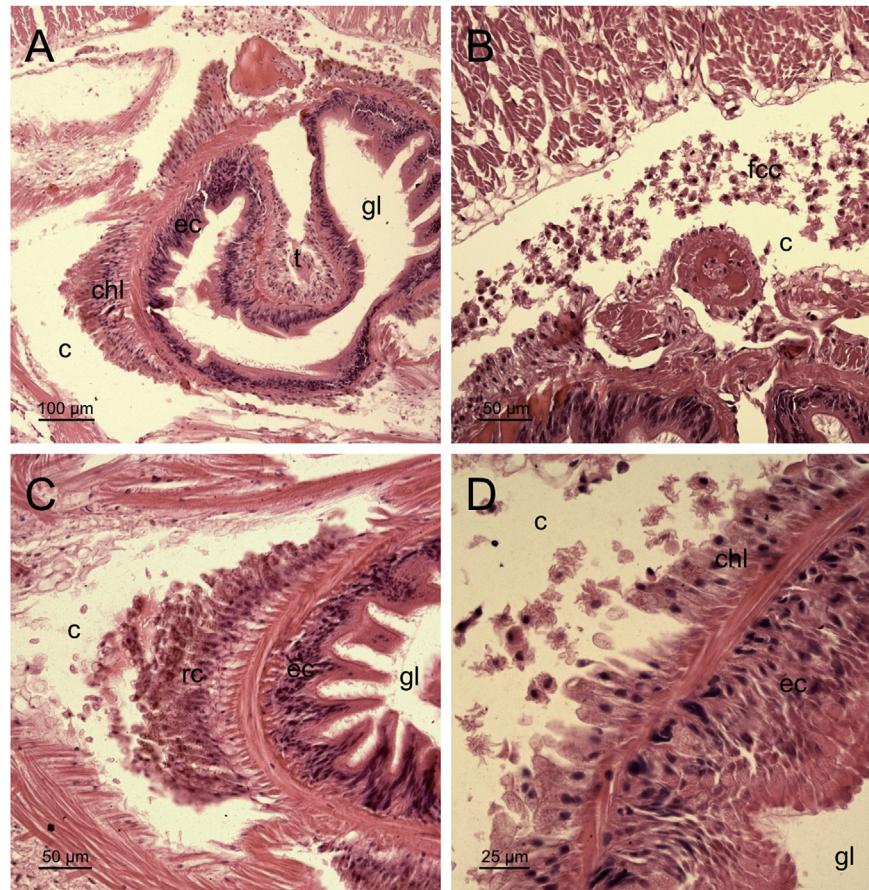


Fig. 4. Transversal sections representing *E. andrei* midgut stained with hematoxylin/eosin. 4A: General view of the structures of the gut. The gut lumen (gl) is lined with a layer of enterocytes (ec). The chloragogen layer (chl) surrounding the tube of the gut is oriented toward the coelom cavity (c). Typhlosolis (t) represents gut invagination that increases the surface for food absorption. 4B: Free-floating coelomocytes (fcc). 4C: Release of coelomocytes (rc) into the coelom from the mesenchymal layer; detail in 4D.

from the mesenchymal coelomic lining.

Based on our results, we propose a model of the earthworm defense against infection. The increased number of microorganisms is sensed by pattern recognition receptors in the gut, and this information is transferred into the adjacent mesenchymal lining representing the precursor tissue of free coelomocytes and the site of their release. The released coelomocytes with immune function act as phagocytes and/or produce antimicrobial proteins and opsonins that reduce the number of microorganisms in the coelomic cavity, thus keeping the infection under control.

Acknowledgments

This research was supported by the Institutional Research Concept RVO 61388971 and by the project “BIOCEV – Biotechnology and Biomedicine Centre of the Academy of Sciences and Charles University in Vestec” (CZ-1.05/1.1.00/02.0109) from the European Regional Development Fund.

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